AJP Policy Document

The environmental impacts of animal-based research

The Animal Justice Party of Aotearoa New Zealand (AJP) aims to ban all harmful forms of animal experimentation. AJP commits to promoting the use of compassionate research.

Animal experimentation often utilises harmful and invasive processes that routinely involve subjecting animals to painful and stressful procedures. The rate at which animal-based models of experiments fail to translate to human-based models is high. This creates an opportunity to find alternative methods that can prove to be more effective, accurate and ethical, for example, there are emerging alternatives to many forms of animal experimentation such as cell cultures and computer simulations. By requiring the use of alternatives to animal testing wherever possible, we can foster a culture of innovation and advancement in the scientific field that does not come at the expense of animals.

However, the traditional, and still dominant, ethical view, which is mirrored in most animal testing legislation and regulations, is that the use of animals in research is acceptable in order to achieve scientific, public safety and medical goals, provided that animal suffering is minimised and that the human benefits gained could not be obtained by utilising alternative methods. By its very nature, animal testing required by law is in tension with the goal of animal welfare legislation, which is to protect animals from violations of their welfare.

Animal-based research isn't just an ethical issue. It also has significant environmental impacts that demand attention. The environmental cost of maintaining these practices contradicts commitments to sustainability and climate action.

The main environmental impacts of animal experimentation include:

1. ***High Resource Consumption***: Animal research facilities use large amounts of water, electricity, and specialised food for the animals. The need to maintain specific

environmental conditions (e.g., temperature control and ventilation) further increases energy use.

Animal research requires significant energy resources. Research facilities consume up to ten times more energy per square metre than office buildings[1] due to stringent requirements, including controlled ventilation, temperature regulation, pathogen barriers, specialised lighting, and power-intensive equipment.

2. ***Waste Production and Pollution***: Research generates substantial biomedical waste that requires careful management and disposal. This includes both biological waste (animal carcasses, excrement, bedding, and uneaten food) and hazardous materials (syringes, needles, contaminated materials, and chemical waste).

   When animals used in research are disposed of, their remains may be classified as hazardous waste, depending on the type of studies they were involved in. This can include exposure to chemicals, infectious agents, or radioactive materials.

   A key concern is the gap in awareness and safety measures between laboratory personnel, who are trained to handle such materials, and external waste management or transportation services, which may not have the same level of knowledge about the risks involved. If these third-party handlers are unaware of potential hazards, improper disposal could lead to environmental contamination or public health risks.

   Proper disposal methods often involve incineration or other resource-intensive processes. During incineration, particulate matter, organic compounds, pathogens, and radioactive materials may be released into the surrounding environment, posing significant human health risks, including cancer and respiratory illnesses. Environmental concerns include air and water pollution, acidification, and greenhouse gas emissions.

3. ***Carbon Footprint***: The transportation of animals, supplies, and research materials contributes to carbon emissions. Additionally, maintaining breeding colonies and research facilities adds to the overall carbon footprint.

4. ***Land Use and Habitat Destruction***: Large-scale breeding and housing of research animals requires land, which could otherwise be used for conservation or more sustainable purposes.

5. ***Chemical and Pharmaceutical Pollution***: Animal testing is particularly used for safety (toxicity) and efficacy testing of chemical compounds. Beyond toxicity testing, animals are also used to assess the effectiveness of sanitation, disinfection, and sterilisation processes.

[1] Animal Free Science Advocacy. (n.d.). *Environmental Impact*
https://animalfreescienceadvocacy.org.au/issues/environmental-impact/

A key concern is that some chemicals used in these experiments may have unknown hazardous or carcinogenic properties, posing risks not only to the animals but also to researchers, waste handlers, and the environment. Many chemicals tested in laboratories may not yet have fully understood toxicological profiles, potentially exposing both animals and humans to unknown dangers.

Industries that conduct toxicity and pharmaceutical testing on animals can contribute to groundwater and soil contamination if their waste management practices are inadequate. Improper disposal of chemical agents, pharmaceuticals, and biological waste can lead to hazardous substances leaching into the environment, causing long-term environmental hazards and public health risks.

Key Objectives:

Interim measures, while we work towards the AJP Policy Ending Animal testing, include:

1. AJP will encourage the minimisation of the environmental impacts of animal-based research by requiring all research facilities, laboratories, and institutions engaged in animal research and testing, including academic, governmental, and private-sector organisations, to implement sustainable practices, reduce waste, and ensure compliance with relevant environmental regulations.

2. AJP will support the adoption of the following measures by all research facilities and institutions:

    - *Reduce, Refine, Replace (3Rs)*: Prioritise alternatives in order to replace animal testing where possible, refine methods to minimise environmental impact and reduce the number of animals used. Strengthen the adherence to the 3Rs.
    - *Waste Management*: Reduce reuse and recycle materials wherever possible, properly dispose of hazardous and biological waste in compliance with regulations and implement sustainable alternatives for animal bedding and feed where feasible.
    - *Energy and Resource Efficiency*: Utilise energy-efficient equipment and lighting in animal research facilities and optimise water usage in cleaning, animal care and laboratory operations.
    - *Sustainable Procurement*: Source laboratory supplies, animal feed and equipment from environmentally responsible suppliers and prioritise biodegradable or recyclable materials in research processes.

- *Reduction of Greenhouse Gas Emissions*: Reduce transportation-related emissions by consolidating supply shipments and using local vendors when possible, implementing practices that minimise the release of harmful substances into the air and water.

References:
Animal Free Science Advocacy. (n.d.) Environmental Impact. https://animalfreescienceadvocacy.org.au/issues/environmental-impact/. Accessed May 2025.
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